Ceramics

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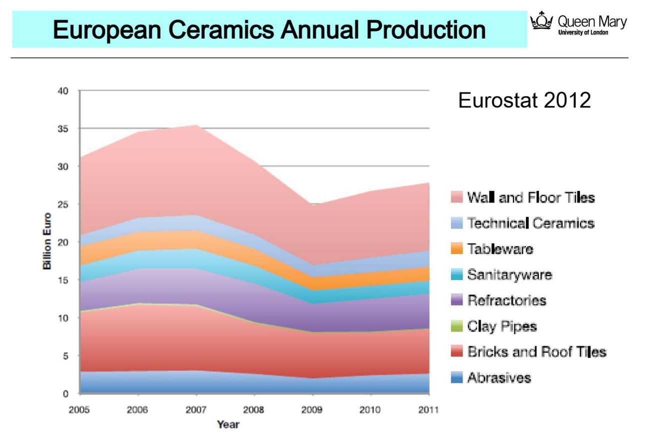
Ceramics:

- Inorganic, non-metallic
- Most are compounds between metallic & non-metallic elements; interatomic bonds → Totally ionic, or mostly ionic with some covalent

Traditional Ceramics:

- Until 1950s, primary raw material was clay
- e.g: China, porcelain, bricks, tiles, glasses, high temp ceramics



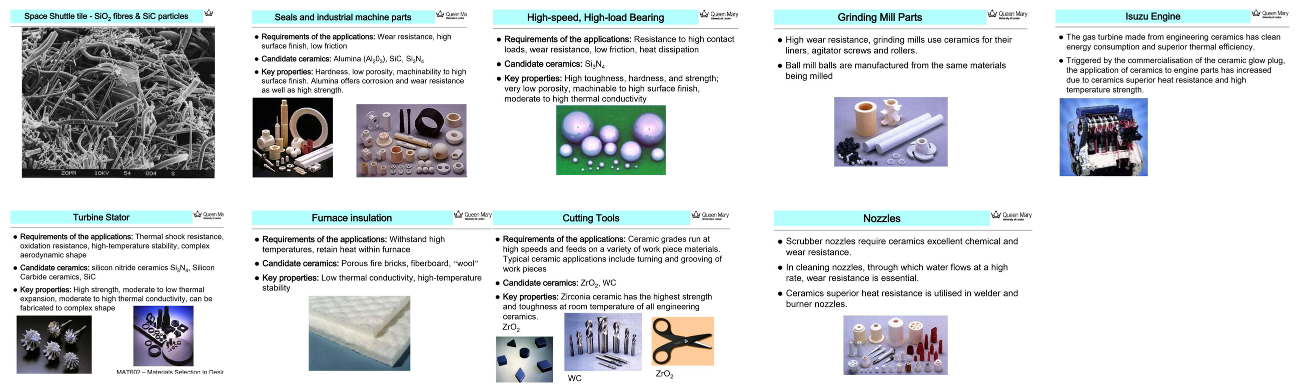
Modern Ceramics:

- Wear and corrosion resistance → Pumps, nozzles, valve parts.
- High heat + thermal shock resistance → Automotive engine applications
- Can be made into extremely precise shapes → Within fraction of micron

Material groups:

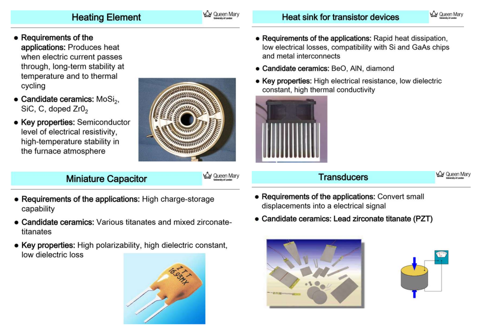
Structural – often high temp applications

e.g: Si_3N_4, SiAlON, Al_2O_3, ZrO_2



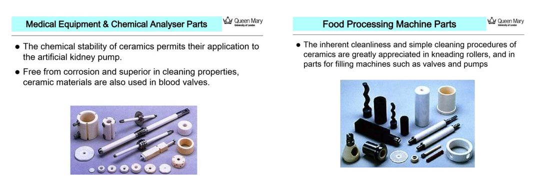
Electrical: Used for electrical / magnetic applications

e.g: PZT, $BaTiO_3$



Bioceramics: Used for medical applications

e.g. Al_2O_3, hydroxyapatite, glass.



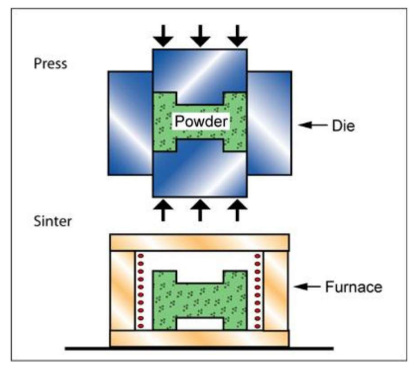
Processing:

Milling – Raw powders milled in distilled water

Drying and granulation – Dry slurry, crush agglomerates, sieve powder

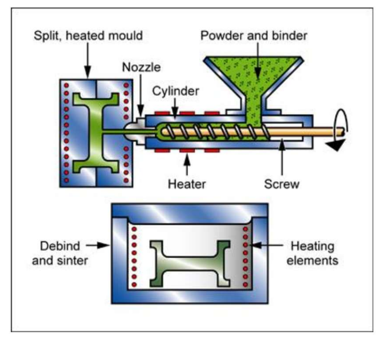
Compaction – Press materials

Sintering: Heat the compact to cause densification



Injection moulding:

- Ceramic powder mixed with organic binder
- Injected using standard injection moulding machine
- Binder removed by thermal baking or using solvents
- Green part can be sintered



Additives:

- Selection depends on:
  - Green strength required
  - Ease of machining
  - Compatability with ceramic powder
  - Nature of consolidation process

- Gums, waxes, thermoplastic resins, and thermosetting resins are not soluble in water and do not provide a benefit for slip casting, but are excellent for the warm mixing used to prepare a powder for injection moulding.
- Organic binders can be burned off at low temperature and result in minimal contamination, whereas inorganic binders become a part of the composition.

Binder	Increase green strength and provide lubrication.
Lubricant	Mold releases, decrease particle-tool and inter-particle friction
Compaction aid	Aid in particle rearrangement during pressing.
Plasticizer	Rheological aid, improving flexibility of binder films.
Deflocculant	pH control, control of surface charge on particles, dispersion, or coagulation.
Wetting agent	Reduction of surface tension.
Water retention aid	Retain water during pressure application.
Antistatic agent	Charge control.
Antifoam agent	Prevent foam or bubble formation.
Foam stabilizer	Enhances foam formation.
Chelating agent	Deactivate undesirable ions.
Fungicide	Stabilize against degradation with aging.
Sintering aids	Aid in densification.
Dopants	Modify electrical, magnetic, optical properties.
Phase stabilizers	Control the crystalline phases present.

Raw materials Selection Criteria:

- Dependant on properties required in finished component
- Purity:
  - Influences properties such as strength, stress, rupture life & oxidation resistance
  - Effect of impurity dependant on chemistry of ceramic and impurity, distribution of impurity, and service conditions of component (time, temp., stress, environment).
    - For instance, Ca decreases the creep resistance of Si_3N_4 hot-pressed with MgO as a densification (sintering) aid, but appears to have little effect on Si_3N_4 hot-pressed with Y_2O_3 as the densification aid.
  - Effect of impurities more important for electrical, magnetic, optical properties. Slight variations in conc. or distribution of dopant severely alter properties.
- Particle size distribution:
  - Consolidation step → Max particle packing & uniformity, so min. shrinkage and retained porosity result during densification.
  - Uniform particle size ≠ good packing
  - Optimum packing for particles all same size = Over 30% Void space
  - Adding particles of size = largest voids, reduces void content to 26%.
  - Adding 3rd still smaller particle leads to 23% pore volume.
  - ∴ For max particle packing, range of particles required
  - Real ceramic particles generally irregular in shape, don't fit in ideal packing.
    - ↳ Porosity after compaction > 50%
  - Low porosity + fine grain size = high strength
  - When strength isn't main concern, large particles + high porosity are important to achieve properties such as low thermal conductivity & high thermal shock resistance

- Mechanical Screening
- Elutriation
- Air classification
- Ball milling
- Attrition milling
- Vibratory milling
- Turbomilling
- Fluid energy milling
- Hammer milling
- Roll crushing
- Chemical Precipitation
- Sol-gel
- Liquid mix
- Decomposition
- Freeze drying
- Hot kerosene drying
- Calcining
- Rotary kiln
- Fluidized bed
- Combustion synthesis

Finer Particle Size

MAT602 – M

- Reactivity:

- Densification of a compacted powder at high temperature is driven by a change in surface free energy.
- Very small particles with high surface area have high surface free energy and thus have a strong thermodynamic drive to decrease their surface area by bonding together. Very small particles, approximately 1μm or less, can be compacted into a porous shape and sintered at a high temperature to near-theoretical density.
- For polycrystalline alumina for sodium vapour lamp envelopes to achieve transparency, virtually all the pores larger than about 0.4μm must be removed during sintering as they scatter the visible wavelengths of light.
- The particle size distribution and reactivity are important in determining the temperature and the time at temperature necessary to achieve sintering.
- Typically, the finer the powder and the greater its surface area, the lower are the temperature and time at temperature for densification.
- This can have an important effect on strength. Long times at temperature result in increased grain growth and lower strength.
- To optimise strength, a powder that can be densified quickly with minimal grain growth is desired.

Sintered Si_3N_4

- Starting powder of approximately 2μm average particle size only sinters to about 90% of theoretical density.
- Sub-micron powder with a surface area roughly greater than 10 m^2/g sinters to greater than 95% of theoretical density.

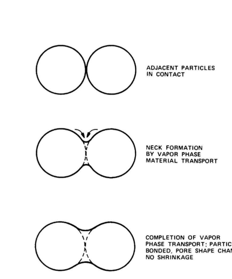
Sintering:

- Pores removed with growth together with strong bonding between adjacent particles
- Mechanism of transport must be present.
- Energy source must be present to activate and sustain material transport process.

Process	Transport Mechanism	Driving Energy
Vapour phase	Evaporation and condensation	Differences in vapour pressure
Solid State	Diffusion	Differences in free energy
Liquid phase	Viscous flow and diffusion	Capillary pressure, surface energy
Reactive liquid	Viscous flow and solution precipitation.	Capillary pressure, surface energy

Vapour Phase Sintering

- Movement from a positive radius of curvature to negative radius with a lower vapour pressure.
- Smaller particles therefore have a stronger driving pressure.
- No shrinkage or densification



Solid State Sintering

Driven by a reduction in free energy of the system.

$$\frac{\Delta L}{L_0} = \left(\frac{K\gamma a^3 D^* t}{kTd^n}\right)^m$$

g = surface energy

a^3 = atomic volume of the diffusing vacancy

D^* = self diffusion coefficient

k = Boltzmann constant

T = temperature

d = particle diameter

t = time

K = geometric constant

n (~3) and m (~0.4) power relationships

Small particles grow more rapidly.